

Design Choices in Retrieval-Augmented Generation: A Systematic Study of Output Factors

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Abstract

Retrieval-Augmented Generation (RAG) enhances Large Language Models (LLMs) by grounding outputs in retrieved external knowledge. However, output reliability and efficiency vary significantly depending on system configuration choices. This study aims to examine the effects of key RAG design choices on factuality, faithfulness, citation accuracy, latency, and computational cost. These choices include document chunking, retrieval depth, reranking, prompting tactics, and decoding settings. The goal is to provide empirically grounded guidance for building robust and efficient RAG pipelines.

Background / Introduction

While LLMs generate fluent responses, they often produce hallucinated or unsupported claims in knowledge-intensive tasks. RAG eliminates this by locating pertinent papers and adding them to the context of generation. The output quality can be greatly impacted by little adjustments to prompting limits, retrieval settings, or segmentation. The fact that most optimization techniques used today are heuristic emphasizes the necessity of a methodical and controlled assessment of these design decisions.

Objectives

- Determine which configuration factors have the biggest effects on output reliability.
- Quantify trade-offs between answer quality, latency, and token cost.
- Analyze interaction effects between selected design parameters.
- Develop practical configuration guidelines for stable RAG performance.

Methods / Methodology

A controlled experimental framework will be employed using a fixed baseline RAG pipeline. Key variables under investigation include chunking strategy, retriever configuration, top-k retrieval depth, reranking mechanisms, grounding prompts, context window size, and decoding parameters. Experiments will be conducted across multiple domain-diverse datasets, including subsets with ground truth answers and retrieval identifiers. Evaluation will use automated “LLM-as-a-judge” techniques and benchmarking tools to measure factuality, faithfulness, context relevance, citation accuracy, latency, and inference cost. Parameter isolation and selected pairwise interaction analyses will also be performed.

Results / Findings

Preliminary observations will suggest that retrieval depth improves recall up to a threshold, beyond which irrelevant context degrades performance. Reranking significantly enhances evidence quality, and semantic chunking improves retrieval precision compared to fixed segmentation. Strict grounding may restrict the scope of expression while reducing unsubstantiated assertions.

Conclusion

This study will establish a reproducible framework for evaluating how RAG configuration choices affect output quality and efficiency. The findings aim to support evidence-based engineering practices for reliable and cost-effective RAG deployment.

Keywords

Retrieval-Augmented Generation; Large Language Models; Information Retrieval; Faithfulness; Factuality; Reranking; Prompt Engineering; Latency Optimization.